Non-Permanent History Framework for Econometrics

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Abstract

Classical econometrics usually treats the past as fixed once observed. This paper develops a non-permanent history framework in which historical data are themselves stochastic objects, indexed by both the reference date of the event and the vintage date at which the event is read. The primitive object is not Y_t but $Y(t, \tau)$, where t denotes historical time and τ denotes the vintage or reading time. The framework introduces historical uncertainty, revision risk, vintage-conditional likelihoods, revision-aware identification, vintage-robust estimation, and non-commuting asymptotics. Applications are developed for economics, finance, political science, geopolitics, international relations, warfare economics, warfare, statistics, algorithms, data science, and artificial intelligence. The central claim is that when histories are non-permanent, econometric inference must become inference over triangular arrays of mutable records rather than fixed sequences of observations.

Contents

List of Figures

List of Tables

1 Introduction

Econometric practice often assumes that the past is known. An observation such as gross domestic product in year t, inflation in quarter t, battlefield losses in month t, or a sovereign risk spread on day t is commonly represented by a scalar Y_t. This representation hides a critical fact: many economically and politically important observations are repeatedly revised. National accounts are revised, intelligence estimates are revised, conflict casualty counts are revised, financial balance sheets are restated, corporate earnings are amended, population figures are rebased, and archival political records are reclassified.

The central object of this paper is therefore not a time series, but a *history-vintage surface*. The value assigned to historical date t at reading date τ is denoted

$$Y(t, \tau), \qquad \tau \geq t.$$

The inequality $\tau \geq t$ expresses the fact that period t can only be read at or after period t. The resulting data object is triangular. Classical econometrics observes one row, one column, or one diagonal of a richer object and then implicitly treats it as permanent.

This paper develops a framework for econometrics when histories are non-permanent. The framework has six goals:

1. define historical observations as vintaged stochastic objects;
2. distinguish real-time data, revised data, and final data;
3. derive the consequences of historical uncertainty for bias, variance, identification, structural breaks, and asymptotics;
4. construct vintage-aware estimators and tests;
5. connect the framework to economics, finance, political science, geopolitics, international relations, warfare economics, warfare, statistics, data science, and AI/ML;
6. provide implementable algorithms for empirical work.

2 The Single Axiom

Axiom 1 (Non-permanence of history)**.** *Histories are not permanent.*

The axiom means that the recorded value of a historical event may change after the event occurs. It does not require metaphysical alteration of the event itself. It requires only that the *recorded, measurable, decision-relevant, and inference-relevant representation* of the event can change across vintages.

Definition 1 (Historical date and vintage date)**.** *The historical date t is the period to which an observation refers. The vintage date τ is the period at which the observation is read, recorded, disseminated, or used for inference.*

Definition 2 (Non-permanent observation). *A non-permanent observation is a map*

$$Y : \{(t, \tau) \in \mathbb{R}^2 : \tau \geq t\} \times \Omega \to \mathbb{R}$$

such that $Y(t, \tau)$ *is measurable with respect to the information available at vintage* τ*.*

The axiom immediately implies that the scalar representation Y_t is generally incomplete. The correct primitive is $Y(t, \tau)$.

3 The Vintage Data Process

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space with a filtration

$$\{\mathcal{F}_\tau : \tau \in T\},$$

where T is the calendar-time index set. The vintage data process is

$$\{Y(t, \tau) : t \in T,\ \tau \in T,\ \tau \geq t\}.$$

Assumption 1 (Adapted vintage process). *For all* $t \leq \tau$, $Y(t, \tau)$ *is* $\mathcal{F}_\tau$*-measurable.*

Definition 3 (Real-time value). *The real-time value of historical period* t *is*

$$Y_t^{RT} = Y(t, t).$$

Definition 4 (Vintage-τ historical sample). *At vintage* τ*, the available historical sample is*

$$\mathcal{Y}_\tau = \{Y(t, \tau) : t \leq \tau\}.$$

Definition 5 (Final value). *If the limit exists in probability, the final value of period* t *is*

$$Y(t, \infty) = \lim_{\tau \to \infty} Y(t, \tau).$$

The final value need not be metaphysical truth. It is the limiting record under a given archival, statistical, political, or institutional process.

4 Triangular Data Geometry

The data matrix is triangular:

	$\tau = 1$	$\tau = 2$	$\tau = 3$	$\cdots$	$\tau = T$
$t = 1$	$Y(1,1)$	$Y(1,2)$	$Y(1,3)$	$\cdots$	$Y(1,T)$
$t = 2$		$Y(2,2)$	$Y(2,3)$	$\cdots$	$Y(2,T)$
$t = 3$			$Y(3,3)$	$\cdots$	$Y(3,T)$
$\vdots$				$\ddots$	$\vdots$
$t = T$					$Y(T,T)$

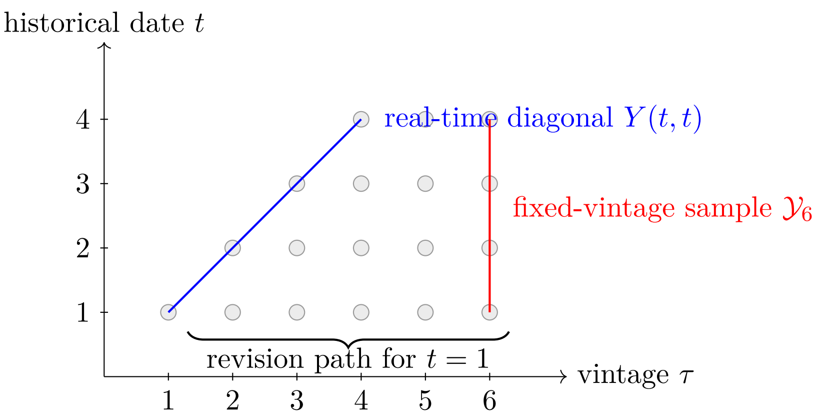

Figure 1: Triangular geometry of non-permanent historical data.

5 Revision Processes

Definition 6 (Revision)**.** *For $\tau_2 \geq \tau_1 \geq t$, the revision of period t between vintages τ_1 and τ_2 is*

$$R(t; \tau_1, \tau_2) = Y(t, \tau_2) - Y(t, \tau_1).$$

A one-period revision is

$$r(t, \tau) = Y(t, \tau) - Y(t, \tau - 1), \qquad \tau > t.$$

Then

$$R(t; \tau_1, \tau_2) = \sum_{s=\tau_1+1}^{\tau_2} r(t, s).$$

Definition 7 (Revision bias)**.** *The revision bias at date t from vintage τ_1 to vintage τ_2 is*

$$b_R(t; \tau_1, \tau_2) = \mathbb{E}[R(t; \tau_1, \tau_2)].$$

Definition 8 (Revision volatility)**.** *The revision volatility is*

$$\sigma_R^2(t; \tau_1, \tau_2) = \mathrm{Var}(R(t; \tau_1, \tau_2)).$$

Definition 9 (Historical uncertainty function)**.** *The historical uncertainty remaining about period t at vintage τ is*

$$H(t \mid \tau) = \mathrm{Var}(Y(t, \infty) \mid \mathcal{F}_\tau).$$

Classical econometrics effectively assumes $H(t \mid \tau) = 0$ for all $t < \tau$. The non-permanent history framework permits $H(t \mid \tau) > 0$.

6 Parametric Revision Models

The revision process can be modeled in several ways.

6.1 Martingale Revision Model

A rational statistical agency model is

$$\mathbb{E}[Y(t, \tau + 1) \mid \mathcal{F}_\tau] = Y(t, \tau).$$

Equivalently,

$$\mathbb{E}[r(t, \tau + 1) \mid \mathcal{F}_\tau] = 0.$$

This says revisions are news, not predictable corrections.

6.2 Mean-Reverting Revision Model

A conservative initial estimate model is

$$Y(t, \tau + 1) - \mu_t = \rho\{Y(t, \tau) - \mu_t\} + \varepsilon_{t,\tau+1}, \qquad |\rho| < 1.$$

Here μ_t may represent the final value, a latent truth, or a slowly changing statistical target.

6.3 Biased Revision Model

A politically or institutionally biased revision model is

$$r(t, \tau + 1) = a_t + c_\tau + z_t'\delta + u_{t,\tau+1},$$

where a_t is a historical-date effect, c_τ is a vintage-date effect, and z_t contains political, institutional, geopolitical, financial, or wartime covariates.

6.4 State-Space Revision Model

Let Y_t^* denote a latent final value and $Y(t, \tau)$ denote noisy vintage measurement:

$$\begin{aligned} Y_t^* &= \phi Y_{t-1}^* + \eta_t, \\ Y(t, \tau) &= Y_t^* + m(t, \tau), \\ m(t, \tau + 1) &= \rho m(t, \tau) + \nu_{t,\tau+1}. \end{aligned}$$

This nests measurement error, real-time macroeconomics, nowcasting, and revision dynamics.

7 Economic Interpretation

Economic decisions are often made using real-time data but evaluated using revised data. Let a policymaker choose action a_t according to

$$a_t = \pi(Y(1, t), Y(2, t), \ldots, Y(t, t), s_t),$$

where s_t is other information at time t. An econometrician at vintage $\tau > t$ estimates the policy rule using

$$Y(1, \tau), Y(2, \tau), \ldots, Y(t, \tau).$$

The econometrician may therefore estimate the decision rule using information the policymaker did not possess.

Definition 10 (Historical information mismatch)**.** *The information mismatch for historical decision date t evaluated at vintage τ is*

$$M(t, \tau) = \mathcal{G}_{t,\tau} - \mathcal{G}_{t,t},$$

where

$$\mathcal{G}_{t,\tau} = \sigma(Y(s, \tau) : s \leq t).$$

This mismatch is central in monetary policy, fiscal policy, financial regulation, sovereign risk, war mobilization, crisis response, and geopolitical intelligence.

8 Finance

In finance, non-permanent history arises in restated earnings, revised balance sheets, reclassified sovereign reserves, modified credit ratings, benchmark revisions, and survivorship bias.

Let P_t be an asset price and let $F(t,\tau)$ be the vintage-τ fundamental for period t. A vintage-aware pricing equation is

$$P_t = \mathbb{E}_t \left[\sum_{j=1}^{\infty} \frac{D(t+j,t)}{(1+r)^j} \right],$$

where dividends, cash flows, or fundamentals are evaluated using real-time vintage information.

A historical researcher instead may estimate

$$P_t = \mathbb{E}_\tau \left[\sum_{j=1}^{\infty} \frac{D(t+j,\tau)}{(1+r)^j} \right].$$

The difference between these two equations is the vintage valuation wedge:

$$VW(t,\tau) = \mathbb{E}_\tau \left[\sum_{j=1}^{\infty} \frac{D(t+j,\tau) - D(t+j,t)}{(1+r)^j} \right].$$

9 Political Science and Geopolitics

Political data are often revised through recounts, archival openings, intelligence disclosures, census rebasing, casualty revisions, classification changes, and regime reinterpretations.

Let S_t denote state capacity, C_t conflict intensity, and A_t alliance credibility. A geopolitical model estimated at vintage τ may be written

$$C(t,\tau) = \alpha + \beta_S S(t,\tau) + \beta_A A(t,\tau) + \varepsilon(t,\tau).$$

The same war or crisis may have different measured causes under different vintages:

$$\widehat{\beta}_S(\tau_1) \neq \widehat{\beta}_S(\tau_2).$$

Definition 11 (Geopolitical revision shock)**.** *A geopolitical revision shock is a change*

$$R_G(t;\tau_1,\tau_2) = G(t,\tau_2) - G(t,\tau_1)$$

in a geopolitical variable G, caused by new intelligence, declassification, territorial reinterpretation, casualty revision, regime change, or archival correction.

Such shocks can change estimated alliance reliability, deterrence credibility, state capacity, and the inferred causes of conflict.

10 Warfare Economics and Warfare

War is an extreme environment for non-permanent history. Battlefield information is incomplete, strategic deception is intentional, casualty numbers are disputed, territorial control changes rapidly, and post-war archives often alter the historical record.

Let $B(t,\tau)$ denote assessed battlefield balance at historical time t and vintage τ. Let $L(t,\tau)$ denote assessed losses. Let $Q(t,\tau)$ denote logistical capacity. A commander at time t chooses operation o_t according to

$$o_t = \pi(B(t,t), L(t,t), Q(t,t), I_t),$$

where I_t is intelligence available at time t.

A historian or military economist at vintage τ estimates

$$o_t = \pi(B(t,\tau), L(t,\tau), Q(t,\tau), I_\tau).$$

The latter may be a poor representation of the decision environment faced by the commander.

Definition 12 (Fog-of-war revision)**.** *A fog-of-war revision is a revision caused by delayed battlefield observation, deception, propaganda, intelligence failure, classification, or post-conflict documentation.*

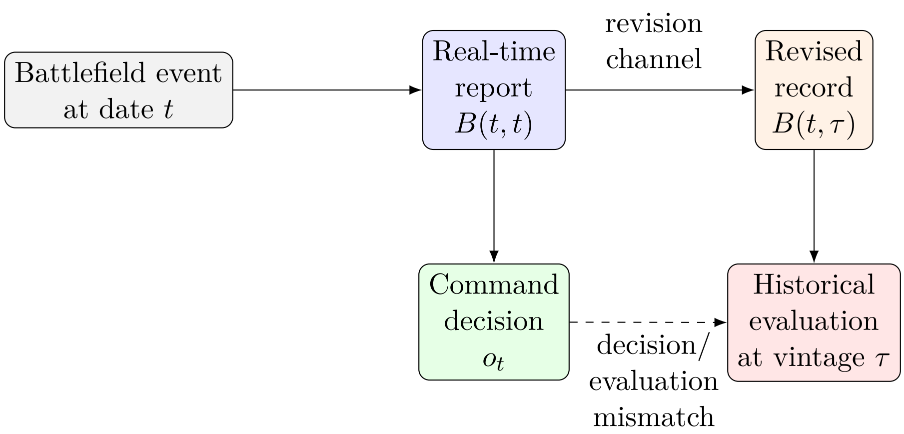

Figure 2: Non-permanent history in warfare: real-time command and revised evaluation.

11 Vintage-Conditional Likelihood

Let $\theta \in \Theta$ be a structural parameter and let $\mathcal{Y}_\tau$ be the historical data available at vintage τ. The vintage-conditional likelihood is

$$L_\tau(\theta) = f(\mathcal{Y}_\tau \mid \theta).$$

The maximum likelihood estimator is

$$\widehat{\theta}_\tau = \arg\max_{\theta\in\Theta} L_\tau(\theta).$$

If $\mathcal{Y}_{\tau_1} \neq \mathcal{Y}_{\tau_2}$, then generally

$$\widehat{\theta}_{\tau_1} \neq \widehat{\theta}_{\tau_2}.$$

Definition 13 (Vintage score)**.** *The vintage score is*

$$S_\tau(\theta) = \frac{\partial}{\partial\theta} \log L_\tau(\theta).$$

Definition 14 (Revision score)**.** *The revision score between vintages τ_1 and τ_2 is*

$$\Delta S(\theta; \tau_1, \tau_2) = S_{\tau_2}(\theta) - S_{\tau_1}(\theta).$$

A model is revision-stable if $\Delta S(\theta; \tau_1, \tau_2)$ is small in probability for all large τ_1, τ_2.

12 OLS Under Non-Permanent History

Consider the linear model

$$y^* = X^*\beta + \varepsilon, \qquad \mathbb{E}[X^{*\prime}\varepsilon] = 0,$$

where y^* and X^* are final or latent values. At vintage τ, the econometrician observes

$$y_\tau = y^* + u_\tau, \qquad X_\tau = X^* + V_\tau.$$

The vintage-τ OLS estimator is

$$\widehat{\beta}_\tau = (X_\tau' X_\tau)^{-1} X_\tau' y_\tau.$$

Theorem 1 (Revision bias in OLS)**.** *Suppose $X_\tau = X^*$ and $y_\tau = y^* + u_\tau$. If $X^{*\prime}X^*$ is nonsingular, then*

$$\mathbb{E}[\widehat{\beta}_\tau \mid X^*] = \beta + (X^{*\prime}X^*)^{-1} X^{*\prime}\mathbb{E}[u_\tau \mid X^*].$$

Hence OLS is conditionally unbiased if and only if

$$X^{*\prime}\mathbb{E}[u_\tau \mid X^*] = 0.$$

Proof. Substitute $y_\tau = X^*\beta + \varepsilon + u_\tau$ into the OLS formula:

$$\widehat{\beta}_\tau = (X^{*\prime}X^*)^{-1} X^{*\prime}(X^*\beta + \varepsilon + u_\tau).$$

Thus

$$\widehat{\beta}_\tau = \beta + (X^{*\prime}X^*)^{-1} X^{*\prime}\varepsilon + (X^{*\prime}X^*)^{-1} X^{*\prime}u_\tau.$$

Taking conditional expectations and using $\mathbb{E}[\varepsilon \mid X^*] = 0$ gives the result. □

Corollary 1 (Generic failure of unbiasedness)**.** *If revisions are systematically related to regressors, then vintage OLS is biased.*

This is especially plausible in macroeconomics, finance, war, and political economy, where revisions are often correlated with recessions, crises, regime changes, wars, data suppression, or institutional capacity.

13 Uncertainty Decomposition

Let $\widehat{Y}_{t+h|\tau}$ be a forecast made at vintage τ for period $t+h$. Define final forecast error

$$e_{t+h|\tau} = Y(t+h, \infty) - \widehat{Y}_{t+h|\tau}.$$

The mean squared forecast error decomposes as

$$\mathbb{E}[e_{t+h|\tau}^2] = U_F + U_H + U_S + U_V + 2C,$$

where:

$$\begin{aligned}
U_F &= \text{future shock uncertainty},\\
U_H &= \text{historical revision uncertainty},\\
U_S &= \text{structural parameter uncertainty},\\
U_V &= \text{vintage path uncertainty},\\
C &= \text{cross-covariance among uncertainty components}.
\end{aligned}$$

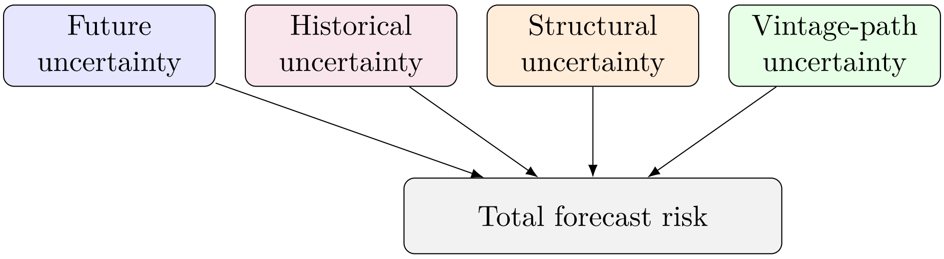

Figure 3: Forecast uncertainty under non-permanent history.

Classical forecast evaluation usually includes U_F and parts of U_S, but ignores U_H and U_V.

14 Structural Breaks and Revision Artifacts

Let the researcher estimate

$$Y(t,\tau) = \begin{cases} x_t'\beta_1 + \varepsilon_t, & t \le t^*, \\ x_t'\beta_2 + \varepsilon_t, & t > t^*. \end{cases}$$

A detected break at t^* may reflect:

1. a genuine change in the data-generating process;
2. a change in measurement technology;
3. a change in political reporting incentives;
4. a change in archival access;
5. a change in statistical definitions;
6. a revision process that is itself nonstationary.

Theorem 2 (Break-revision indeterminacy)**.** *For any observed vintage-τ series $\{Y(t,\tau)\}_{t=1}^T$ exhibiting an apparent structural break at t^*, there exists a no-break latent series $\{Y_t^*\}_{t=1}^T$ and a revision process $\{u(t,\tau)\}_{t=1}^T$ such that*

$$Y(t,\tau) = Y_t^* + u(t,\tau),$$

and the apparent break is entirely generated by $u(t,\tau)$.

Proof. Let the observed series be $Y(t,\tau)$. Choose any no-break latent process, for example

$$Y_t^* = x_t'\beta + \eta_t$$

with constant β. Define the revision component by

$$u(t,\tau) = Y(t,\tau) - Y_t^*.$$

Then the identity $Y(t,\tau) = Y_t^* + u(t,\tau)$ holds exactly. Since Y_t^* has no structural break by construction, all break behavior in the observed series is assigned to the revision component. Therefore, without restrictions on $u(t,\tau)$, the observed break is not separately identified from revision dynamics. □

Corollary 2 (Need for revision restrictions)**.** *Structural breaks are identified only under auxiliary restrictions on the revision process, such as mean-zero revisions, stationarity of revisions, external validation data, institutional metadata, or revision-orthogonal instruments.*

15 Vintage Consistency

Definition 15 (Sample consistency)**.** *An estimator $\widehat{\theta}_{n,\tau}$ is sample-consistent at fixed vintage τ if*

$$\operatorname*{plim}_{n\to\infty} \widehat{\theta}_{n,\tau} = \theta_\tau.$$

Definition 16 (Vintage consistency)**.** *An estimator $\widehat{\theta}_{n,\tau}$ is vintage-consistent if*

$$\operatorname*{plim}_{\tau\to\infty} \operatorname*{plim}_{n\to\infty} \widehat{\theta}_{n,\tau} = \theta^*.$$

The two limits

$$\lim_{n\to\infty} \lim_{\tau\to\infty} \widehat{\theta}_{n,\tau} \qquad \text{and} \qquad \lim_{\tau\to\infty} \lim_{n\to\infty} \widehat{\theta}_{n,\tau}$$

need not commute.

Theorem 3 (Non-commuting vintage asymptotics)**.** *There exist revision processes for which*

$$\lim_{n\to\infty} \lim_{\tau\to\infty} \widehat{\theta}_{n,\tau} \neq \lim_{\tau\to\infty} \lim_{n\to\infty} \widehat{\theta}_{n,\tau}.$$

Proof. Let

$$\widehat{\theta}_{n,\tau} = \theta^* + \frac{1}{n} + a_\tau,$$

where a_τ is a vintage distortion. If $a_\tau \to 0$, then both limits coincide. But define

$$a_\tau = \begin{cases} 0, & \tau < n, \\ 1, & \tau \geq n. \end{cases}$$

Along paths where n grows before τ, the distortion can vanish; along paths where τ dominates n, it can persist. More generally, if $a_{\tau,n}$ depends on the relative path of (n, τ), the two iterated limits need not coincide. □

16 Identification Under Revision Volatility

Let

$$Y(t,\tau) = X_t'\beta^* + \varepsilon_t + u(t,\tau).$$

If $u(t,\tau)$ is unrestricted, then β^* is not identified. Any alternative parameter $\widetilde{\beta}$ can be rationalized by

$$\widetilde{u}(t,\tau) = u(t,\tau) + X_t'(\beta^* - \widetilde{\beta}).$$

Theorem 4 (Non-identification)**.** *Without restrictions on revision errors, β^* is not identified from vintage data alone.*

Proof. The observed value is

$$Y(t,\tau) = X_t'\beta^* + \varepsilon_t + u(t,\tau).$$

For any $\widetilde{\beta}$, define

$$\widetilde{u}(t,\tau) = u(t,\tau) + X_t'(\beta^* - \widetilde{\beta}).$$

Then

$$X_t'\widetilde{\beta} + \varepsilon_t + \widetilde{u}(t,\tau) = X_t'\beta^* + \varepsilon_t + u(t,\tau) = Y(t,\tau).$$

Thus the same observed data are compatible with infinitely many structural parameters. □

Identification requires additional assumptions, such as:

$$\mathbb{E}[u(t,\tau) \mid X_t] = 0,$$

or valid instruments Z_t satisfying

$$\mathbb{E}[Z_t u(t,\tau)] = 0.$$

17 Vintage-Robust Estimation

17.1 Vintage-Robust OLS

Let

$$H_\tau = \mathrm{diag}(H(1 \mid \tau), \dots, H(n \mid \tau)).$$

Define weights

$$w_t(\tau) = \frac{1}{H(t \mid \tau) + \epsilon},$$

where $\epsilon > 0$ prevents division by zero. Let

$$W_\tau = \mathrm{diag}(w_1(\tau), \dots, w_n(\tau)).$$

The vintage-robust OLS estimator is

$$\widehat{\beta}_{VR} = (X_\tau' W_\tau X_\tau)^{-1} X_\tau' W_\tau y_\tau.$$

Proposition 1 (Reduction to OLS)**.** *If $H(t \mid \tau) = h$ is constant for all t, then $\widehat{\beta}_{VR}$ equals OLS.*

Proof. If $W_\tau = (h+\epsilon)^{-1} I$, then

$$(X' W_\tau X)^{-1} X' W_\tau y = \{(h+\epsilon)^{-1} X'X\}^{-1} (h+\epsilon)^{-1} X'y = (X'X)^{-1} X'y.$$

□

17.2 Revision-Augmented VAR

Let

$$Z_t(\tau) = \begin{pmatrix} Y(t,\tau) \\ R(t; \tau-1, \tau) \end{pmatrix}.$$

A revision-augmented VAR is

$$Z_t(\tau) = A_1 Z_{t-1}(\tau) + \cdots + A_p Z_{t-p}(\tau) + e_t(\tau).$$

This permits impulse responses to be decomposed into:

1. economic shocks;
2. measurement shocks;
3. institutional revision shocks;
4. vintage-path shocks.

17.3 Doubly-Robust Historical Estimator

Let $m(X_t) = \mathbb{E}[u(t,\tau) \mid X_t]$ be a revision-bias model. Define

$$\widetilde{y}_t(\tau) = Y(t,\tau) - \widehat{m}(X_t).$$

Then

$$\widehat{\beta}_{DR} = (X' W_\tau X)^{-1} X' W_\tau \widetilde{y}.$$

This estimator is designed to be robust if either the weighting model $H(t \mid \tau)$ or the revision-bias model $m(X_t)$ is correctly specified.

17.4 Real-Time Matching Estimator

For treatment $D_t \in \{0, 1\}$, the real-time treatment effect is

$$\tau_{RT} = \mathbb{E}[Y(1, t, t) - Y(0, t, t) \mid X_t].$$

A vintage-matched estimator compares treated and control units using the same vintage structure:

$$\widehat{\tau}_{RTME} = \frac{1}{N_1} \sum_{i:D_i=1} Y_i(t_i, t_i) - \frac{1}{N_0} \sum_{i:D_i=0} Y_i(t_i, t_i).$$

17.5 Vintage-Path Instrumental Variables

Classical IV requires

$$\mathrm{Cov}(Z, \varepsilon) = 0.$$

Vintage-path IV requires

$$\mathrm{Cov}(Z, R(t; \tau_1, \tau_2)) = 0 \qquad \text{for all} \qquad \tau_2 \geq \tau_1 \geq t.$$

The instrument must be orthogonal not only to the structural disturbance but also to revision disturbances.

18 Algorithms

Algorithm 1: Constructing a Vintage Panel

Step	Operation
1	Collect all releases of the target variable.
2	For each release vintage τ, store all historical observations $Y(t, \tau)$ with $t \leq \tau$.
3	Construct the triangular matrix with rows indexed by historical date t and columns by vintage τ.
4	Compute revisions $R(t; \tau_1, \tau_2) = Y(t, \tau_2) - Y(t, \tau_1)$.
5	Estimate historical uncertainty $H(t \mid \tau)$ using revision histories, state-space filtering, bootstrap, or Bayesian posterior variance.
6	Estimate structural models using both real-time and revised samples.
7	Report sensitivity of coefficients, standard errors, forecasts, and tests across vintages.

Algorithm 2: Vintage-Robust OLS

Step	Operation
1	Input y_τ, X_τ, and estimated $H(t \mid \tau)$ for $t = 1, \ldots, n$.
2	Choose small $\epsilon > 0$.
3	Compute weights $w_t = 1/(H(t \mid \tau) + \epsilon)$.
4	Form $W = \mathrm{diag}(w_1, \ldots, w_n)$.
5	Compute $\widehat{\beta}_{VR} = (X'WX)^{-1}X'Wy$.
6	Estimate variance using sandwich or bootstrap procedures that resample both observations and revision histories.
7	Report coefficient estimates with historical-uncertainty-adjusted confidence intervals.

Algorithm 3: Break-Revision Diagnostic

Step	Operation
1	Estimate structural break dates using final or current vintage data.
2	Re-estimate break dates across earlier vintages.
3	Compute the vintage stability score $S_B = \frac{1}{K}\sum_{k=1}^{K} \mathbf{1}\{\widehat{t}^*_{\tau_k} = \widehat{t}^*_{\tau_K}\}$.
4	Estimate whether revision magnitudes increase near the detected break date.
5	Test whether revisions predict the break statistic.
6	Classify the break as stable, revision-sensitive, or revision-generated.

19 Statistical Tests

19.1 Martingale Revision Test

The null hypothesis is

$$H_0 : \mathbb{E}[r(t, \tau + 1) \mid \mathcal{F}_\tau] = 0.$$

A regression test is

$$r(t, \tau + 1) = \alpha + \gamma' Z(t, \tau) + e(t, \tau + 1),$$

where $Z(t, \tau)$ contains vintage-τ predictors. Under the martingale null,

$$\alpha = 0, \qquad \gamma = 0.$$

19.2 Revision Endogeneity Test

For structural regressors X_t, test

$$H_0 : \mathrm{Cov}(X_t, R(t; \tau_1, \tau_2)) = 0.$$

A rejection implies that revisions are not orthogonal to the regressors and that classical estimates are potentially biased.

19.3 Vintage Stability Test

For estimates $\widehat{\beta}_{\tau_1}, \ldots, \widehat{\beta}_{\tau_K}$, define

$$V_\beta = \frac{1}{K}\sum_{k=1}^{K} (\widehat{\beta}_{\tau_k} - \bar{\beta})(\widehat{\beta}_{\tau_k} - \bar{\beta})',$$

where

$$\bar{\beta} = \frac{1}{K}\sum_{k=1}^{K} \widehat{\beta}_{\tau_k}.$$

Large eigenvalues of V_β indicate vintage instability.

20 Data Science and AI/ML

Machine learning models trained on revised data and deployed on real-time data face a form of distribution shift:

$$P_{\text{train}}(X, Y) = P(X(t, \tau_{\text{final}}), Y(t, \tau_{\text{final}})),$$

while

$$P_{\text{deploy}}(X, Y) = P(X(t, t), Y(t, t)).$$

This is a *vintage covariate shift*.

Definition 17 (Vintage covariate shift)**.** *Vintage covariate shift occurs when*

$$P(X(t, \tau_{train})) \neq P(X(t, \tau_{deploy})).$$

Definition 18 (Vintage label shift)**.** *Vintage label shift occurs when*

$$P(Y(t, \tau_{train})) \neq P(Y(t, \tau_{deploy})).$$

Definition 19 (Vintage concept drift)**.** *Vintage concept drift occurs when*

$$P(Y(t, \tau) \mid X(t, \tau))$$

changes with τ even for a fixed historical t.

A vintage-aware empirical risk minimization problem is

$$\widehat{f} = \arg\min_{f \in \mathcal{F}} \sum_{t=1}^{n} w_t(\tau)\ell(f(X(t, \tau)), Y(t, \tau)),$$

where $w_t(\tau)$ penalizes uncertain or unstable historical observations.

21 Reinforcement Learning Under Non-Permanent History

In policy environments, agents act using real-time states but researchers may train using revised states. Let the real-time state be

$$s_t^{RT} = S(t, t),$$

and the revised state be

$$s_t^{RV} = S(t, \tau), \qquad \tau > t.$$

A reinforcement learning policy trained on revised history solves

$$\max_{\pi} \mathbb{E} \left[\sum_{t=0}^{T} \delta^t r(S(t, \tau), a_t) \right],$$

but deployment requires

$$\max_{\pi} \mathbb{E} \left[\sum_{t=0}^{T} \delta^t r(S(t, t), a_t) \right].$$

The resulting policy gap is

$$PG(\pi, \tau) = V^{\pi}(S(t, \tau)) - V^{\pi}(S(t, t)).$$

This matters for central banking, fiscal crisis management, wartime command, cyber defense, and supply-chain intervention.

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22 Reporting Standards

A non-permanent history regression table should include vintage information.

Item	Symbol	Meaning
Historical sample	$t = 1, \ldots, n$	Periods to which the observations refer
Data vintage	τ	Date at which data were read or downloaded
Real-time flag	RT	Whether the data correspond to decision-time information
Revision window	$[\tau_1, \tau_2]$	Vintage interval over which revisions are measured
Historical uncertainty	$H(t \mid \tau)$	Remaining uncertainty about final values
Revision bias	b_R	Expected revision between vintages
Vintage stability	V_β	Variation of estimates across vintages
Break stability	S_B	Stability of detected break dates across vintages

Table 1: Minimum reporting standard for non-permanent history econometrics.

23 Comparison with Classical Approaches

Issue	Classical econometrics	Non-permanent history framework
Observation	Y_t	$Y(t, \tau)$
Past uncertainty	Usually zero	Generally positive
Data matrix	Vector or rectangle	Triangular vintage array
Likelihood	$L(\theta \mid Y)$	$L_\tau(\theta \mid \mathcal{Y}_\tau)$
Breaks	DGP changes	DGP or revision changes
Forecast error	Future uncertainty	Future, historical, structural, vintage risk
Replication	Same code, current data	Same code, original vintage
Machine learning risk	Temporal shift	Temporal and vintage shift
Policy evaluation	Ex-post data	Decision-time data

Table 2: Classical versus non-permanent history econometrics.

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24 Applications

Field	Non-permanent object	Consequence
Macroeconomics	GDP revisions	Real-time policy rules differ from ex-post estimates
Finance	Earnings restatements	Historical fundamentals are vintage-dependent
Political science	Election and census revisions	Regime and representation measures shift
Geopolitics	Intelligence revisions	Threat assessments change across archives
International relations	Alliance credibility records	Deterrence models become vintage-sensitive
Warfare economics	Casualty and logistics revisions	Cost-benefit analysis changes after conflict
Warfare	Fog-of-war reports	Command decisions differ from historical reconstructions
Statistics	Mutable observations	Sampling theory must include historical uncertainty
Data science	Dataset drift by vintage	Train-test mismatch across data releases
AI/ML	Revised labels and features	Vintage covariate and label shift

Table 3: Cross-disciplinary implications.

25 Policy Evaluation

Suppose a policy is chosen at time t:

$$D_t = d(Y(t,t), X(t,t), I_t).$$

The ex-post econometrician estimates

$$Y(t+h,\tau) = \alpha + \delta D_t + X(t,\tau)'\beta + \varepsilon_t.$$

But if

$$X(t,\tau) \neq X(t,t),$$

then the control variables may not represent the decision environment. The estimated treatment effect δ may answer the wrong question.

The real-time estimand is

$$\delta_{RT} = \mathbb{E}[Y(1,t+h,t+h) - Y(0,t+h,t+h) \mid \mathcal{F}_t].$$

The revised-history estimand is

$$\delta_{RV} = \mathbb{E}[Y(1,t+h,\tau) - Y(0,t+h,\tau) \mid \mathcal{F}_\tau].$$

Generally,

$$\delta_{RT} \neq \delta_{RV}.$$

26 Replication

Under non-permanent history, replication has two distinct meanings.

Definition 20 (Code replication)**.** *Code replication holds when the same code applied to the same dataset produces the same results.*

Definition 21 (Vintage replication)**.** *Vintage replication holds when the same code applied to the dataset available at the original publication vintage produces the original results.*

Definition 22 (Current-vintage replication)**.** *Current-vintage replication holds when the same code applied to the latest vintage produces the same substantive conclusions.*

A failed current-vintage replication may be a successful discovery of historical revision rather than a failure of the original paper.

27 A General Estimation Problem

Let the researcher minimize a vintage-aware loss:

$$\widehat{\theta}_\tau = \arg\min_{\theta \in \Theta} \sum_{t=1}^{n} \ell(Y(t,\tau), X(t,\tau), \theta) + \lambda \sum_{t=1}^{n} H(t \mid \tau) q(Y(t,\tau), X(t,\tau), \theta).$$

The first term fits the observed vintage data. The second term penalizes reliance on historically unstable observations. The tuning parameter $\lambda \geq 0$ determines how strongly the researcher penalizes historical uncertainty.

28 Practical Workflow

A complete empirical workflow is:

1. Define the target historical variable and collect all available vintages.
2. Construct $Y(t,\tau)$ for all feasible (t,τ).
3. Estimate revision processes and historical uncertainty.
4. Compare real-time, current-vintage, and final-vintage estimates.
5. Test revision martingality and revision endogeneity.
6. Estimate structural models with vintage-robust procedures.
7. Report coefficient, forecast, and policy sensitivity across vintages.
8. Archive the vintage used for publication.

29 Limitations

The framework is demanding. It requires vintage metadata, archival discipline, and sometimes unavailable historical releases. It can also overcomplicate settings where revisions are tiny, random, and irrelevant. The framework is most valuable when:

1. data are heavily revised;
2. decisions were made in real time;

3. policy evaluation matters;
4. structural breaks are central;
5. political, institutional, or wartime incentives affect measurement;
6. ML systems are trained on revised data but deployed on real-time data.

30 Conclusion

The non-permanent history framework begins from a simple axiom: histories are not permanent. From this axiom follows a substantial reconstruction of econometrics. Observations become vintaged objects, parameters become doubly indexed, the past carries uncertainty, structural breaks may be revision artifacts, and classical asymptotics must be supplemented by vintage asymptotics.

The framework is not a rejection of classical econometrics. It is an extension of econometrics to settings where the historical record is mutable. In macroeconomics, finance, political science, geopolitics, international relations, warfare economics, warfare, statistics, data science, and AI/ML, many important datasets are not fixed records but evolving archives. Econometric inference must therefore account not only for uncertainty about the future, but also for uncertainty about the past.

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Glossary

Historical date
The time period to which an observation refers.

Vintage date
The time period at which an observation is read, released, recorded, or used for inference.

Non-permanent history
A historical record whose measured value can change across vintages.

Vintage data process
The two-index process $\{Y(t,\tau) : \tau \geq t\}$.

Real-time data
Data available at the same time as the decision or event being studied.

Final data
The limiting or latest available value of a historical observation.

Revision
A change in the recorded value of a historical observation across vintages.

Historical uncertainty
Conditional uncertainty about the eventual final value of a past observation.

Revision bias
The expected value of a revision.

Revision volatility
The variance of a revision.

Vintage consistency
Convergence of an estimator as the vintage process matures.

Vintage covariate shift
A difference between feature distributions across data vintages.

Vintage label shift
A difference between label distributions across data vintages.

Break-revision indeterminacy
The inability to distinguish structural breaks from revision artifacts without restrictions on the revision process.

Fog-of-war revision
Revision caused by delayed, distorted, classified, or strategically manipulated battlefield information.

Vintage-path IV
Instrumental variables estimation requiring orthogonality to structural shocks and revision shocks.

The End